MILITARY SPECIFICATION

METER, LOW-RANGE RADIAC SURVEY

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Low-Range Radiac Survey Meter, lightweight, shipboard, portable equipment, hereinafter referred to as the equipment, for detecting, indicating, and computing gamma radiation, and for indicating the presence of beta radiation.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-1/906	Electron Tube, Radiation Counter, Type 5979
MIL-E-1/973	Electron Tube, Radiation Counter, Type 5980
MIL-B-18	Battery, Dry
MIL-M-10304/18	Meter, Multiscale, DC, Panel Type, Ruggedized (Round Flange, 3-1/2 Inch, Style 36)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection by Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

DRAWINGS

MILITARY

Oak Ridge National Laboratory

Q-5510-1 through -21	Low-Range Radiac, Probe And Parts List

NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)

RE-D2681324	Cord, Probe, Retractile
RE-F2688037	H-43B/U Headset Assembly

PUBLICATIONS

NATIONAL BUREAU OF STANDARDS (NBS)

NBS 250	Calibration And Test Services Of The National Bureau of Standards

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with all provisions of MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.3).

3.2 Radiacmeter. The radiacmeter shall be capable of detecting and measuring gamma radiation in the intensity range up to 1000 milliroentgens per hour (mR/hr) subject to further requirements herein, and shall be capable of detecting the presence of beta radiation. The radiacmeter shall indicate the presence of 10 millicuries of krypton -85 on the 0 to 10 mR/hr range, with beta shield removed, and shall indicate at least half scale.

3.2.1 Component parts. The equipment shall consist of the following component parts required to make up a complete equipment:

a. One radiacmeter, including probe and probe cable, but not including batteries
b. One high-impedance headset, Type H-43B/U, in accordance with Drawing (DWG) RE-F2688037
c. One carrying strap
d. One technical manual
e. One carrying case

3.2.2 Electronics. Solid-state circuitry shall be employed unless otherwise approved by the procuring activity.

3.2.3 Parts. All parts used in the construction of the equipment shall be selected and applied in accordance with MIL-E-16400. No parts shall be used which require special selection

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from commercial production runs in order to conform to the equipment requirements. No mercury or mercury compound shall be used in this equipment. Parts shall be rated at 125° Celsius (C) or higher if required by circuit application. Plastic encapsulated microcircuits shall not be used, unless approved by the procuring activity. Germanium semiconductor devices shall not be used.

3.2.3.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3 Physical requirements.

3.3.1 Size and weight. The size and weight of the following components shall be no greater than specified herein:

a. Radiacmeter:
   1. Length, 11.50 inches (in)
   2. Width, 5.00 in
   3. Height, 8.50 in
   4. Weight, 7.50 pounds (lbs), not including batteries
b. Probe:
   1. Length, 7.375 in
   2. Width, 1.75 in
   3. Height, 1.59375 in
   4. Weight, 1.1 lbs
c. Total weight: Total weight shall not exceed 12.0 lbs, with technical manual, not including batteries.

3.3.2 Radiacmeter housing. The radiacmeter housing shall be a watertight enclosure and shall contain the electronic circuitry (except the probe), meter, and batteries. The material of the housing shall be #356 aluminum.

3.3.2.1 Radiacmeter housing construction. The housing shall be of split-type gasketed construction, with the seam at the top, and held closed by captive screws inserted from the top surface. The top shall act as a panel to support the indicating meter and the electronic chassis. The top shall also include an integral compartment for the batteries. The compartment shall be accessible only through a separate cover in the upper surface, held by captive screws. The bottom of the housing shall be a cover which shall enclose the electronic chassis and the battery compartment.

3.3.2.2 Radiacmeter finish. Unless otherwise specified in the contract (see 6.2), the radiacmeter housing shall be treated against corrosion and painted gray in accordance with MIL-E-16400. The surface shall be as smooth as possible with the minimum number of crevices to collect contaminated material. Rubber or felt feet shall not be permitted. The surface shall be easily decontaminable without recourse to laboratory methods.

3.3.2.3 Calibration access cover. A cover shall be provided in the top panel to protect the calibration controls. The cover shall not compromise the watertightness of the enclosure.

3.3.2.4 Probe well. On the front end of the exterior of the housing there shall be a receptacle for holding the probe in a vertical position. The receptacle shall be integral with the lower part of the housing.

3.3.2.5 Detector cable. The cable between the radiacmeter housing and the probe shall be retractile type in accordance with DWG RE-D2681324. Strain relief devices shall be provided at the points of attachment to the probe and radiacmeter.

3.3.3 Detector (probe). The radiacmeter shall be equipped with a detector (probe) which shall house two Geiger-Muller (GM) tubes, Types 5980 and 5979 (see 3.7.2), covering ranges of 0 to 1000 mR/hr and 0 to 1 mR/hr, respectively. The probe shall consist of one smooth-contoured cylindrical metal housing, one for both tubes. There shall be a movable beta-ray shield opposite the beta window of the type 5979. The overall outside diameter of the housing shall not be more than 1.60 in for a length of 7.375 in overall, and the housing (for both tubes) shall project not

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more than 0.178 in beyond the end of the 5979 tube. The detector shielding shall be so mounted as to permit ready access to, and replacement of, the tubes. The degree of enclosure of the probe shall be watertight in accordance with MIL-STD-108. Markings indicating the center of sensitivity of each tube and its orientation shall be etched into the housing surface. Material and finish of the probe housing shall be as specified in Oak Ridge DWG Q-5510-17.

3.3.4 Handles. A fold-down handle shall be attached to the radiacmeter housing. The carrying case shall be provided with at least one handle for transporting purposes.

3.3.5 Carrying case. The equipment shall include a carrying case, adequately compartmented, to store and protect the following components:

a. Radiacmeter
b. Headset
c. Carrying strap (nomenclature to be assigned) (harness)
d. One technical manual
e. One set of spare batteries

3.3.5.1 Material. The material of the carrying case shall be drawn aluminum, as light as possible consistent with the strength required.

3.3.5.2 Degree of enclosure. The degree of enclosure shall be splash-proof in accordance with MIL-STD-108.

3.3.5.3 Finish. The surface of the carrying case shall be treated against corrosion and painted gray in accordance with MIL-E-16400. The surface shall be as smooth as possible with the minimum number of crevices to collect contaminated material.

3.3.6 Headset assembly. One headset, high-impedance, Type H-43B/U, shall be included in the equipment. A receptacle shall be provided for connection of the headset to the equipment. Material and finish of the headset shall be as specified in DWG RE-F2688037.

3.3.7 Carrying strap. There shall be one carrying strap, or shoulder harness, included in the equipment. The strap shall provide for carrying the radiacmeter in either the hip or abdominal position, and shall be attachable via a latch to the designed points on the top of the radiacmeter housing. When not in use, it shall be stored in the carrying case. Material and finish of the carrying strap shall be as specified in Oak Ridge DWG Q-5510-21.

3.3.8 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein.

3.4 Power.

3.4.1 Electrical design. The power source shall be two batteries, Type BA-30 (D-cells), in accordance with MIL-B-18. Batteries are not required to be furnished with the set.

3.4.2 Power supply. A power supply shall be furnished as part of the radiacmeter to provide all necessary voltages for operation of the detector, radiacmeter, and headset.

3.4.2.1 Stability. The power consumption shall be limited to a value such that the radiacmeter will operate continuously for 40 hours or more under normal operating conditions within the stated accuracy limits on one set of batteries. Battery end point shall be 0.9 volts (V) per cell. The 40 hours include eight hours with illuminating lamp on.

3.4.3 Illumination. A push-to-operate switch shall be included on the top of the panel to control the meter illumination lamp. The switch shall be located adjacent to the carrying handle so that the operator may hold the radiacmeter by the handle and operate the switch with the same hand. The lamp shall illuminate the meter sufficiently to permit reading-after-dark adaptation. The lamp shall be an incandescent type, with optics, if necessary, to direct light where required.

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3.4.4 External connectors. External connectors shall be in accordance with MIL-E-16400 and as specified in 3.4.4.1.

3.4.4.1 Test points. Test points shall be identified, but shall not be brought out to the case. Blown fuse indicators are not required.

3.4.5 Indicating meter. The indicating meter shall be a 6-position ruggedized multi-range scale shift microammeter, 50 microamperes (μA) full scale, Type MR36M201SPECR in accordance with MIL-M-10304/18. The meter shall be mechanically coupled to the selector switch so that the scale positions and switch positions will correspond. The first position of the meter shall be common to positions 1 and 2 of the selector switch. The meter shall be located at the top of the radiacmeter housing so as to be visible from above the radiacmeter through a window, whether the radiacmeter is carried by the hand, by the handle, or is carried in the abdominal or hip position by the strap. The meter shall be sealed to the housing to maintain the watertight degree of enclosure.

3.5 Controls.

3.5.1 Operating controls. The operating controls shall consist of a selector switch and a switch to operate the illuminating lamp. All controls shall be classified as continually required, except calibration controls.

3.5.2 Calibration controls. A separate control shall be provided for calibration of each scale range. Access to the controls shall be restricted to prevent tampering.

3.6 Operating characteristics.

3.6.1 Radiacmeter composition. The radiacmeter shall include the following subassemblies and circuitry:

a. Power supply subassembly for regulated 715 V positive and regulated 6.3 V positive
b. Stable electronic pulse rate computer and integrating circuit
c. Detector probe with connecting cable
d. One multi-range indicating meter assembly
e. One control switch and range selector assembly
f. One headset connection and dust-tight cover
g. One lamp for night illumination, with control by switch only
h. Housing to contain and support these and associated components

3.6.2 Radiacmeter operation. The radiacmeter operation shall be as follows:

a. Accuracy. Within the environmental limits specified herein, when the radiacmeter is exposed to gamma radiation whose true value is between 10 and 100 percent of the scale range selected, the reading shall not deviate more that ±20 percent from the true value. For this determination, the radiation shall be incident normal to the plane formed by the major axes of two Geiger-Muller tubes.
b. Calibration. Calibration shall be performed with a gamma source traceable to the NBS. Calibration equipment and procedure shall be approved in advance by the procuring activity.
c. Calibration controls. A separate control shall be provided for calibration of each scale range. Access to the controls shall be restricted to prevent tampering.
d. Saturation. When the radiacmeter is exposed to gamma radiation up to 600 roentgens per hour (R/hr) on any scale, the meter pointer shall indicate off-scale at the high end of the range.
e. Energy response. The response to gamma radiation at any given intensity shall vary not more that ±20 percent throughout the energy range from 0.08 million electron volts (MeV) to 1.33 MeV. Below 0.08 MeV, the relative response shall decrease with decreasing energies.

3.6.3 Selector switch. The control for the selector switch shall be located on the top of the radiacmeter housing and shall be rotatable to even positions to provide for the modes of operation listed. This switch shall be coupled to the meter so that the scale corresponding to the

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switch position is presented.

a. Position 1: OFF. Maximum counterclockwise. The switch shall open at least the minimum number of circuits to ensure zero current drain from the battery. Meter scale: OFF - BATT (same as for b)
b. Position 2: BATTERY CONDITION. The switch shall connect the meter into the circuit in such a way as to indicate the condition of the battery under load equal to the circuit load. Meter scale: OFF - BATT (same as for a)
c. Position 3: 0 to 1000 mR/hr. Meter scale: 0 to 1000
d. Position 4: 0 to 100 mR/hr. Meter scale: 0 to 100
e. Position 5: 0 to 10 mR/hr. Meter scale: 0 to 10
f. Position 6: 0 to 1.0 mR/hr. Meter scale: 0 to 1.0
g. Position 7: 0 to 0.1 mR/hr. Meter scale: 0 to 0.1

3.6.4 Pulse rate computer integrator function. The computer integrator shall receive pulses from the detector at the input and shall furnish at the output an electrical current which shall be directly proportional to the average value of the rate of arrival of pulses, but independent of their amplitude and duration. The computer integrator shall drive a 50- μA indicating meter. An additional output shall present signals representing individual pulses to drive a headset.

3.6.5 Detector-computer-integrator response. The radiacmeter shall measure the intensity of gamma radiation between 0 and 1000 mR/hr under the following conditions:

a. Circuit time constant. For an output current sufficient to produce a deflection of 20 percent or more of full-scale reading, the circuit time constants shall be such that the meter pointer shall reach 90 percent of true indication in not less than 3 seconds (s) and not more than 7 s.
b. Noise. No noise or spurious counts shall be introduced, except from operation of the selector switch. This necessary zero background shall be provided as the result of proper care in design, construction, and choice of parts.

3.7 Design requirements.

3.7.1 Component construction. The radiacmeter, probe, and ancillary components shall be constructed in accordance with the Military drawings referenced in 2.1, revision level R0, dated 1 August 1981.

3.7.2 Special components. The following components are specified for use in design of the equipment:

a. Indicating meter, Type MR36M201SPECR in accordance with MIL-M-10304/18
b. GM tube Type 5979 in accordance with MIL-E-1/906
c. GM tube Type 5980 in accordance with MIL-E-1/973
d. Cord, probe in accordance with DWG RE-D2681324
e. Headset assembly, Type H-43B/U in accordance with DWG RE-F2688037

3.7.3 Environmental conditions. The equipment shall conform to the temperature requirements of MIL-E-16400 for temperature Range 3 (from -40°C to +50°C ±2°C), the storage temperature Range 3 (from -62°C to +71°C).

3.7.4 Shock. The equipment shall conform to the shock requirement specified in MIL-E-16400. After each blow, the equipment shall continue to operate satisfactorily without internal adjustments and without replacement of any part. There shall be no evidence of collision between any part of the equipment.

3.7.5 Vibration. The equipment shall conform to the vibration requirement specified in MIL-E-16400. The instrument shall operate satisfactorily without internal adjustments and without replacement of any part. There shall be no evidence of collision between any parts of the equipment.

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3.7.6 Energy response. The energy response of the radiacmeter shall be determined by irradiation with gamma radiation from cobalt-60 (or cesium-137), and with X-rays at approximately 200, 160, 120, 80, 50, and 30 kilo-electron volts (keV). Tube voltages and filtrations for the particular energies shall be as specified by the NBS for a 250 kilovolts peak (kVP) machine. The response shall fall within the limits specified in 3.6.2 e.

3.7.7 Electromagnetic interference (EMI). Electromagnetic compatibility (EMC) for the equipment shall conform to the requirements of MIL-STD-461, Class A4, tests REO1, REO2, RSO1, RSO2, and RSO3. The limits for test RSO3, shall be as specified in TABLE I in lieu of the RSO3 limits specified in MIL-STD-461.

TABLE I. Frequency and field intensity for EMC.

Frequency (megahertz (MHz))	Field intensity volts per meter (V/m)
0.014 to 2.0	10 average
2.0 to 30 1/	50 average
30.0 to 1000 1/	10 average
1000 to 12,400	5 average

1/ Test at only three discrete frequencies with these ranges

3.7.8 Altitude. The equipment shall be designed to conform to the altitude test specified in 4.5.5.

3.8 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1000 hours.

3.9 Quantitative maintainability requirements. The equipment including Government-furnished material or property (GFM or GFP) (see 6.4) shall have a mean-time-to-repair (MTTR), not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile) not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit boards (PCBs)) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.10 Workmanship.

3.10.1 Workmanship screen. The equipment shall withstand a defect detection vibration screen of random type vibration at 0.04 g^2/hertz (Hz) ±3 decibels (dB) from 80 Hz to 350 Hz and temperature cycling as specified in 4.5.6.1.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not

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limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
   4. In-process inspection (see 4.4.4)
c. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2), at least three units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process	Group A	Group B	Group C
Surface examination	3.1	4.5.1	X		X		
Weight	3.3.1	4.5.1	X		X		
Size	3.3.1	4.5.1	X				
Parts and materials	3.2.3, 3.3.3, 3.3.5, 3.3.6, 3.4.5	4.5.1	X				
Finish	3.3.2, 3.3.5.3	4.5.1	X				
Safety	3.1	4.5.1	X				
General workmanship	3.1	4.5.6	X	X	X		X
Workmanship screen	3.10.1	4.5.6.1	X	X	X		X
Pre-performance	3.1	4.5.1	X			X	
Performance	3.4.2.1, 3.4.3, 3.5, 3.6, 3.7.6	4.5.1	X		X		
Low temperature	3.7.3	4.5.1	X				X
High temperature	3.7.3	4.5.1	X				X
Humidity	3.1	4.5.1	X				X
Salt fog	3.1	4.5.1	X				X
Shock	3.7.4	4.5.1	X				X
Vibration	3.7.5	4.5.1	X				X
Inclination	3.1	4.5.1	X				X
Enclosure	3.3.3, 3.3.5.2	4.5.1	X			X	
Magnetic	3.1	4.5.1	X				
Power	3.4.2	4.5.2	X		X	X	

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TABLE II. Examinations and tests. (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process	Group A	Group B	Group C
EMI	3.7.7	4.5.3	X				X
Altitude	3.7.8	4.5.5	X				
Reliability test	3.8	4.6	X				X
Maintainability demonstration	3.9	4.7	X				X
Beta response	3.2	4.5.1	X		X		
Accuracy	3.6.2	4.5.1	X		X		
Energy response	3.6.2, 3.7.6	4.5.4	X				
Saturation	3.6.2	4.5.1	X		X		
Circuit time constant	3.6.5 a	4.5.1	X		X		
Noise	3.6.5 b	4.5.1	X		X		X

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be made on every equipment offered for delivery. This inspection shall comprise such examination and testing which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in Group A of TABLE II.

4.4.2 Production control inspection (Group B). Production control inspection, including sampling, shall conform to the Group B examinations and tests of TABLE II and to the sampling procedures of MIL-STD-105, using the special inspection level S-3 for normal, tightened, and reduced inspection. The acceptable quality level (AQL) shall be 4.0 percent. This inspection shall consist of the examination and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall be performed on equipments that have passed production inspection.

4.3.3 Environmental inspection (Group C). Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipments that have passed production inspection and shall consist of the examinations and tests specified in Group C of TABLE II.

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected in accordance with the following:

a. One sample from the first 25 production units fabricated, or fraction thereof, produced
b. One sample from each successive 100 units, or fraction thereof, produced

4.4.4 In-process inspections. Each equipment produced shall be subjected to the inspections specified in 4.5.6 through 4.5.6.1.2.

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4.5 Test methods.

4.5.1 Examinations and tests. The examinations and tests specified in TABLE II shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.2 through 4.5.6.1.2.

4.5.2 Power test. The battery test and the power ON-OFF switch shall be incorporated in the meter selection switch. The test shall be conducted with fresh batteries as specified in 3.4.1 to demonstrate the requirements of 3.4.2.1. Accuracy shall be within the limits specified in 3.6.2 a at the completion of this test. Current drain shall be established for use as a production test measurement.

4.5.3 EMI tests. EMI test shall be performed on equipments that have passed production inspection and shall consist of test RS03 as specified in 3.7.8 and TABLE I. One sample from each successive 100 units produced, or fraction thereof, shall be tested.

4.5.4 Energy response test. The energy response of the radiacmeter shall be determined by irradiation with cobalt-60, cesium-137, and with X-rays at the effective energies listed in TABLE III. Tube voltages and filtrations for the effective energies shall be as specified in TABLE III. (See NBS Special Publication #250 of 1968 for further information regarding a 250 kVP machine.) For this test, sources of cobalt-60 and cesium-137 should irradiate the radiacmeter probe assembly directly through the sources' encapsulation. No other external shielding shall be interposed between these sources and the probe assembly. The response shall fall within the limits specified in 3.7.7. The response to these X-ray energies shall be as specified in TABLE III.

TABLE III. X-ray machine filtration.

Step	kVP	Equivalent inherent filter millimeters (mm) Al	Added filter (mm)				Approximate effective energy	Required response 1/
			Pb	Sn	Cu	Al		
A	50	4	0.12	0.0	0.0	0.0	40	0.10 - 0.35
B	60	4	0.0	0.0	0.82	0.0	50	0.30 - 0.55
C	100	4	0.53	0.0	0.0	0.0	70	0.60 - 1.00
D	150	4	0.0	1.5	4.0	0.0	120	0.80 - 1.20
E	200	4	0.7	4.0	0.6	0.0	170	0.80 - 1.20
F	250	4	2.7	1.0	0.6	0.0	215	0.80 - 1.20

1/ Required response is the ratio of the instrument measured exposure rate to the true exposure rate at the given effective X-ray energy.

4.5.5 Altitude. The equipment shall actively monitor a source for a period of 24 hours while exposed to an air pressure of 8.8 inches of mercury (in Hg) at a temperature of -40°C, and shall then be turned off and exposed to an air pressure of 3.4 in Hg at a temperature of -54°C for an additional 23-hour period. The equipment shall conform to the performance requirements of this specification throughout the first 24-hour period and subsequent to this test.

4.5.6 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.10 and 3.10.1. Each solder connection and its associated wiring or leads shall be visually examined.

4.5.6.1 Workmanship screen.

4.5.6.1.1 Vibration. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions

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of FIGURE 1. The control accelerometer shall be located next to one of the mounting points of the item under vibration. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The item shall be energized during vibration and exposed to appropriate radiation sources to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

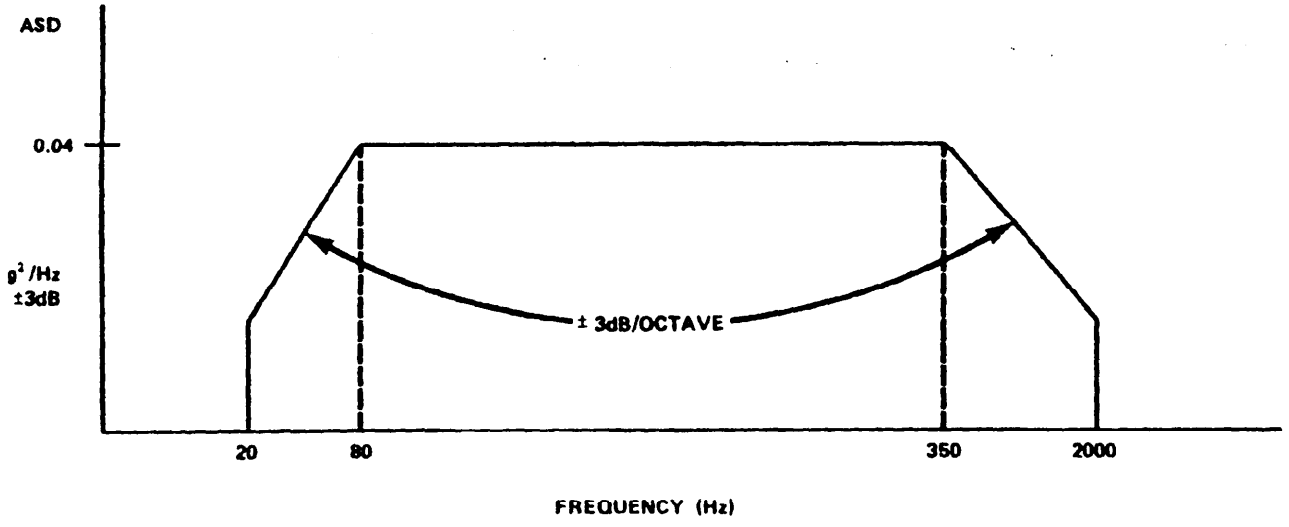


FIGURE 1. Random vibration curve.

4.5.6.1.2 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve specified in FIGURE 2. The number of cycles required shall be determined by the electrical and electronic parts count and the parts count chart (see TABLE IV). The temperature rate of change shall not be less than 5°C per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

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TABLE IV. Electrical and electronic parts count.

Electrical and electronic parts count	Number of temperature cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

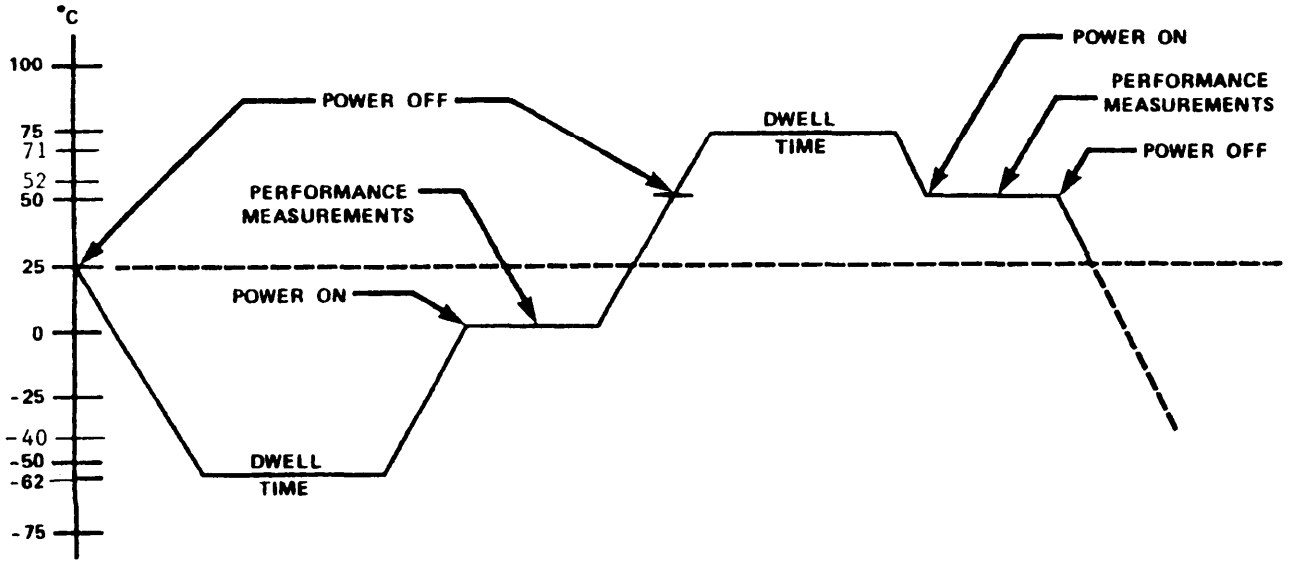


FIGURE 2. One cycle of temperature curve (not to scale).

4.6 Reliability test. Reliability tests shall be performed on randomly selected equipments in accordance with the general and detailed requirements of MIL-STD-781 to determine compliance with the quantitative requirements of 3.8. Testing shall be performed on each production lot beginning with the first lot produced after award of the production contract. Testing shall proceed to an accept or reject decision in accordance with the criteria specified by Test Plan VIII C of MIL-STD-781 for Category 3A equipment.

4.6.1 Failures. The final decision as to the relevancy or nonrelevancy of a failure shall be made by the procuring activity using the failure categories criteria of MIL-STD-781 as a guide. Any operating discrepancy or degradation that requires on unscheduled adjustment or calibration to be made after initial satisfactory operation of the equipment under test shall be considered a relevant failure.

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4.6.2 Rejected equipment. In the event that the equipment is rejected as a result of the reliability test or when pattern failures occur the contractor shall promptly notify the procuring activity and develop corrective action in accordance with the Corrective action paragraph of MIL-STD-781. The whole lot from which the test sample was drawn shall be modified as required, and the reliability test shall be repeated using another random sample drawn from the same lot.

4.6.3 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be reconditioned as required and subjected to the Group C production tests to ensure that it conforms to the requirements of this specification. Subsequently, the equipment may be delivered as part of the contract quantity provided it conforms to all terms and conditions of the criteria for acceptance.

4.7 Maintainability demonstration.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment quantitative maintainability requirements are fulfilled.

4.7.2 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Forty candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471). The following information shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity, or its authorized representative, shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.7.3 Accept or reject criteria. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE V.

TABLE V. Acceptance criteria.

	MTTR	M_{maxct}
Acceptance level	8	0
Sample size	20	20

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (MTTR, M_{maxct}) is less than or equal to that specified in TABLE V corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.7.4 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

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4.7.5 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.6 Rejection. Failure to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies, and
b. Upon approval of an approach to the redesign or correction, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

4.7.7 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to any delivery.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to verify the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preservation, packaging, packing, and marking shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use onboard ships and in shore facilities.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date fo this specification
b. Radiacmeter finish if other than specified in 3.3.2.2
c. Number of first article samples to be submitted if other than specified in 4.3
d. Applicable levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be first production items. The first article should consist of three units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 GFM and GFP. The contracting officer should arrange to furnish the GFM and GFP (see 3.9).

Preparing activity:
NAVY-EC
(Project 6665-N431)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-28864(EC), Meter, Low-Range Radiac Survey

NAME OF ORGANIZATION AND ADDRESS
Naval Electronic Systems Command
(ELEX-8111)
Washington, DC 20360

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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